Inter-Annotator Agreement Analysis Framework: Practical Implementation for Hierarchical Label Assessment

Technical Report

September 8, 2025

Abstract

This report presents the mathematical framework underlying the implemented inter-annotator agreement (IAA) analysis system for hierarchical labeling tasks. The framework encompasses three core analysis dimensions: overall agreement assessment using Krippendorff's α, frequency-stratified agreement analysis, and hierarchical facet comparisons. The implementation uses bootstrap confidence intervals and percentage agreement measures to provide comprehensive reliability assessment for annotation tasks.

1 Introduction and Notation

Let $D = \{d_1, d_2, \ldots, d_n\}$ represent a set of n documents to be annotated, and $A = \{a_1, a_2, \ldots, a_m\}$ represent a set of m annotators. For hierarchical labeling tasks, each annotation consists of a tuple (L_1, L_2) where L_1 represents the parent category and L_2 represents the child category.

1.1 Data Structure

The annotation data can be represented as a matrix $X \in \mathbb{R}^{m \times n}$ where:

$$X_{ij} = \text{label assigned by annotator } a_i \text{ to document } d_j \tag{1}$$

For hierarchical analysis, we define three label spaces:

$$\mathcal{L}_{\text{full}} = \{(l_1, l_2) : l_1 \in \mathcal{L}_1, l_2 \in \mathcal{L}_2\} \tag{2}$$

$$\mathcal{L}_1 = \{\text{parent categories}\} \tag{3}$$

$$\mathcal{L}_2 = \{\text{child categories}\} \tag{4}$$

2 Krippendorff's Alpha: Chance-Corrected Agreement

2.1 General Formulation

Krippendorff's α is defined as:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{5}$$

where D_o is the observed disagreement and D_e is the expected disagreement under the assumption of independence.

2.2 Implementation Algorithm

Algorithm 1 Krippendorff's Alpha Calculation (as implemented)

Require: Reliability data matrix $X \in \mathbb{R}^{m \times n}$
Ensure: α value

```
Encode categorical labels using LabelEncoder
Create pivot table: annotators × documents
Convert to reliability_data array format

return α = krippendorff.alpha(reliability_data, level_of_measurement='nominal')
```

3 Percentage Agreement Metrics

3.1 Overall Percentage Agreement

The overall percentage agreement across all documents is defined as:

$$P_{\text{overall}} = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}_{\{x_{1j}=x_{2j}=\cdots=x_{mj}\}} \tag{6}$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function that equals 1 when all annotators agree on document j.

3.2 Pairwise Agreement Matrix

For annotators a_i and a_k, the pairwise agreement is:

$$P_{ik} = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}_{\{x_{ij}=x_{kj}\}} \tag{7}$$

The complete pairwise agreement matrix is:

$$\mathbf{P} = \begin{pmatrix} 100 & P_{12} & \cdots & P_{1m} \\ P_{21} & 100 & \cdots & P_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m1} & P_{m2} & \cdots & 100 \end{pmatrix} \tag{8}$$

where diagonal elements are 100% and off-diagonal elements are percentage agreements.

3.3 Document-Level Agreement Analysis

For each document d_j, we calculate:

- Perfect agreement indicator: $\mathbf{1}_{\{\text{all annotators agree on } d_j\}}$
- Number of unique labels assigned to d_j
- Number of annotators who labeled d_j

4 Frequency-Based Stratification Analysis

4.1 Label Frequency Distribution

Let f_ℓ denote the frequency of label ℓ across all annotations:

$$f_\ell = \sum_{i=1}^{m} \sum_{j=1}^{n} \mathbf{1}_{\{x_{ij}=\ell\}} \tag{9}$$

4.2 Frequency-Based Stratification

Using quantiles Q_{33} and Q_{67} (33rd and 67th percentiles), we partition labels into three frequency strata:

$$\mathcal{S}_{\text{rare}} = \{\ell : f_\ell \leq Q_{33}\} \tag{10}$$

$$\mathcal{S}_{\text{moderate}} = \{\ell : Q_{33} < f_\ell \leq Q_{67}\} \tag{11}$$

$$\mathcal{S}_{\text{common}} = \{\ell : f_\ell > Q_{67}\} \tag{12}$$

4.3 Stratified Agreement Analysis

For each stratum $\mathcal{S}_s$, we create a subset of data containing only annotations with labels in that stratum:

$$X^{(s)} = \{x_{ij} : x_{ij} \in \mathcal{S}_s\} \tag{13}$$

We then calculate stratum-specific agreement:

$$\alpha^{(s)} = \text{KrippendorffAlpha}(X^{(s)}) \tag{14}$$

If Krippendorff's α fails due to insufficient variation, we use simple agreement rate as fallback:

$$\text{SimpleAgreement}^{(s)} = \frac{\text{Number of documents with perfect agreement in } \mathcal{S}_s}{\text{Total documents in } \mathcal{S}_s} \tag{15}$$

5 Hierarchical Analysis

5.1 Three-Level Decomposition

For hierarchical labels (L_1, L_2), we analyze agreement at three levels:

5.1.1 Parent Level Analysis (L1)

Extract parent labels: $X_{ij}^{(1)} = \pi_1(X_{ij})$ where π_1 projects to the first component.

5.1.2 Child Level Analysis (L2)

Extract child labels: $X_{ij}^{(2)} = \pi_2(X_{ij})$ where π_2 projects to the second component.

5.1.3 Full Hierarchical Analysis

Use complete label tuples: $X_{ij}^{(\text{full})} = X_{ij} = (L_1, L_2)$ concatenated as strings.

5.2 Conditional Agreement Analysis

For each parent category $\ell_1 \in \mathcal{L}_1$, we analyze agreement on child categories within that parent:

$$\text{ConditionalAnalysis}(\ell_1) = \text{Agreement analysis on } \{x_{ij} : \pi_1(x_{ij}) = \ell_1\} \tag{16}$$

5.3 Parent-Child Combination Analysis

We calculate agreement rates for specific (L_1, L_2) combinations by computing perfect agreement rates for documents where at least one annotator used that specific combination.

6 Bootstrap Confidence Intervals

For Krippendorff's α, we construct confidence intervals using bootstrap resampling at the document level:

Algorithm 2 Bootstrap Confidence Interval (as implemented)

Require: Data matrix X, confidence level $(1-\gamma)$, bootstrap samples B
Ensure: Confidence interval $[L, U]$

```
for b = 1 to B do
    Sample documents with replacement: D^(b) ~ Uniform(D)
    Filter annotations for sampled documents
    IF |D^(b)| > 10 THEN
    Compute α^(b) = KrippendorffAlpha(X^(b))
    IF α^(b) is not NaN THEN add to bootstrap collection
    ENDIF
end for
L = percentile(γ/2 × 100)
U = percentile((1 − γ/2) × 100)

return [L, U]
```

7 Implementation Considerations

7.1 Computational Complexity

The implemented algorithms have the following computational complexities:

- Krippendorff's α: $O(mn \cdot |\mathcal{L}|^2)$ where $|\mathcal{L}|$ is the number of unique labels
- Percentage agreement: $O(mn)$
- Pairwise matrix: $O(m^2 n)$
- Bootstrap intervals: $O(B \cdot \text{base computation})$ where B is the number of bootstrap samples
- Hierarchical analysis: $O(3 \cdot mn \cdot |\mathcal{L}|^2)$ for three label types
- Frequency stratification: $O(mn + |\mathcal{L}| \log |\mathcal{L}|)$ for sorting

7.2 Numerical Stability and Error Handling

The implementation handles several edge cases:

- **Insufficient data**: Minimum sample size checks ($n > 10$ documents)
- **Krippendorff failure**: Fallback to simple agreement rate calculation
- **Missing annotations**: Automatic exclusion from pairwise comparisons using pandas dropna()
- **Bootstrap failures**: Skip invalid bootstrap samples and report valid sample count
- **Single-label strata**: Return perfect agreement (1.0) when all labels in stratum are identical
- **Empty intersections**: Return NaN for undefined agreement calculations

7.3 Data Preparation

1. **Label encoding**: Categorical labels converted to numeric using sklearn LabelEncoder
2. **Pivot transformation**: Data reshaped to annotator $\times$ document matrix format
3. **Missing value handling**: NaN values preserved in reliability data matrix
4. **Type conversion**: Arrays converted to float64 for numerical stability

8 Practical Interpretation Guidelines

8.1 Krippendorff's Alpha Interpretation

- $\alpha \geq 0.8$: Excellent reliability
- $0.67 \leq \alpha < 0.8$: Good reliability (acceptable for tentative conclusions)
- $0.4 \leq \alpha < 0.67$: Fair reliability (questionable)
- $\alpha < 0.4$: Poor reliability (unacceptable)

8.2 Confidence Interval Interpretation

Confidence intervals that:

- Do not include 0.67: Strong evidence for reliability level
- Cross reliability thresholds: Uncertain reliability classification
- Are very wide (> 0.2): Insufficient data for reliable estimation

8.3 Hierarchical Analysis Insights

- $\alpha^{(1)} > \alpha^{(2)} > \alpha^{(\text{full})}$: Expected pattern (easier to agree on broad categories)
- $\alpha^{(\text{full})} \approx \alpha^{(1)}$: Child categories may be redundant
- $\alpha^{(2)} > \alpha^{(1)}$: Unexpected pattern requiring investigation

9 Dashboard Integration

The mathematical framework is implemented across three main dashboard components:

1. **Overall Agreement Analysis**: Krippendorff's α with bootstrap CI, pairwise matrices, document-level statistics
2. **Frequency Analysis**: Stratified analysis by label frequency with distribution plots
3. **Hierarchical Analysis**: Three-level decomposition with conditional and combination analysis

Each component provides both numerical results and interactive visualizations for practical decision-making.

10 Conclusions

This framework provides a comprehensive foundation for IAA analysis that balances theoretical rigor with practical implementation:

1. **Chance-corrected reliability**: Krippendorff's α accounts for expected agreement by chance
2. **Intuitive interpretation**: Percentage agreements complement α with easily interpretable metrics
3. **Robust confidence estimation**: Bootstrap methods provide reliable confidence intervals
4. **Hierarchical insights**: Multi-level decomposition reveals structure-specific agreement patterns
5. **Frequency effects**: Stratification analysis identifies how label rarity affects agreement
6. **Error resilience**: Comprehensive error handling ensures stable operation

The framework supports reliable assessment of annotation quality for complex hierarchical labeling tasks while maintaining computational efficiency and numerical stability.